

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 7

Tutorial exercises

1. Suppose we prepare a particle in the ground state of the infinite square well of length a . If we gradually move the right wall out to $2a$, then
 - (a) Write the final wavefunction according to the adiabatic theorem.
 - (b) Write the final wavefunction according to the sudden approximation.
 - (c) Write the condition for the validity of these approximations and analyze.
2. A particle of mass m is inside a box of length L and in its n th state (energy E_n). On a time scale $\tau \ll \hbar/E_n$, the length of the box is increased to $3L$. At later times, what is the probability of finding it in an eigenstate with energy E_n ?
3. Consider a particle confined in a one-dimensional infinite square well where the right boundary moves outward with constant velocity v .
 - (a) Find the wavefunction at time t using adiabatic approximation.
 - (b) From the expression deduce the validity of adiabatic approximation.
 - (c) Find the time scale for the internal dynamics.

Practice questions:

4. A particle is in the ground state of a harmonic oscillator with frequency ω . The frequency is suddenly changed to 2ω . What is the probability that the system remains in the ground state?
5. A particle in the ground state of a box of length L undergoes two sudden changes: first expanding to $2L$ then shrinking back to L . What is the probability that it returns to the original ground state?
6. Consider a spin-1/2 particle in a time-dependent magnetic field of constant magnitude whose direction rotates slowly with angular velocity ω in time. The Hamiltonian is given by $H(t) = -\gamma \vec{B}(t) \cdot \vec{S}$ with $\vec{B}(t) = B_0(\sin \theta \cos \omega t, \sin \theta \sin \omega t, \cos \theta)$

Assuming the system is initially in an instantaneous eigenstate of the Hamiltonian:

- (a) Use the adiabatic approximation to determine the time evolution of the state vector
- (b) Show explicitly that the system remains in the same instantaneous eigenstate
- (c) Hence show that the transition probability to the other eigenstate is zero.